

Talent Scheduling with Daily Capacity and Changeover Times:

A Pattern-Based Integer Linear Programming Approach

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Introduction

In film production:

- Scenes are generally not filmed in the order they appear in the final version of the movie.
- The order of scenes can be arranged based on logistic and economic factors.

The Talent Scheduling (TS) problem concerns finding an optimal filming schedule that minimizes the total holding cost of actors:

- initially considered in [1] and shown to be strongly NP-hard.
- analyzed in [2] by introducing daily working capacity and of changeover times between scenes,

We follow the second approach, which allows multiple scenes to be filmed in a day and accounts for makeup/set adjustments between each scene.

Setup

Consider $m \in \mathbb{Z}^+$ actors and $n \in \mathbb{Z}^+$ scenes and define the following parameters:

- $\mathbf{c} = (c_i)_{m \times 1}$, the vector containing the daily cost of each actor
- $T = (t_{i,j})_{m \times n}$, the matrix representing which actors are in which scenes
- $\mathbf{l} = (\ell_j)_{n \times 1}$, the vector indicating the duration of each scene
- $S = (s_{i,j})_{m \times n}$, the matrix containing the changeover time from scene i to j
- D , the number of working days
- W , the daily working capacity

Feasible Patterns

A subset $P_k \subseteq \{1, \dots, n\}$ is called a **feasible pattern** if there exists an ordering $(j_1, j_2, \dots, j_r)$ of scenes satisfying the daily capacity constraint:

$$\sum_{p=1}^r \ell_{j_p} + \sum_{p=1}^{r-1} s_{j_p, j_{p+1}} \leq W$$

Since changeover times are asymmetric, the feasibility of a subset depends on the existence of at least one valid ordering of its scenes. We define $\mathcal{P} = \{P_1, P_2, \dots, P_{|\mathcal{P}|}\}$ as the collection of all feasible patterns.

Mathematical Model

The TS problem can be modeled using Integer Linear Programming (ILP), where we want to minimize the total actor holding costs by assigning a feasible scene subset $P_k \in \mathcal{P}$ to each day. Let

$$a_{k,j} = \begin{cases} 1 & \text{if scene } j \text{ is in } P_k \\ 0 & \text{otherwise} \end{cases}$$

Similarly, define the following binary decision variables:

- $x_{k,d} = 1$ if pattern P_k is assigned to day d
- $y_{i,d_1,d_2} = 1$ if actor i is held from day d_1 to day d_2

The problem can thus be written as

$$\min \sum_{i=1}^m c_i \sum_{d_1=1}^D \sum_{d_2=d_1}^D (d_2 - d_1 + 1) y_{i,d_1,d_2} \quad (1)$$

$$\text{s.t.} \quad \sum_{k=1}^{|\mathcal{P}|} x_{k,1} = 1, \quad (2)$$

$$\sum_{k=1}^{|\mathcal{P}|} x_{k,d-1} \geq \sum_{k=1}^{|\mathcal{P}|} x_{k,d}, \quad 2 \leq d \leq D, \quad (3)$$

$$\sum_{d=1}^D \sum_{k=1}^{|\mathcal{P}|} a_{k,j} x_{k,d} = 1, \quad 1 \leq j \leq n, \quad (4)$$

$$\sum_{d_1=1}^D \sum_{d_2=d_1}^D y_{i,d_1,d_2} = 1, \quad 1 \leq i \leq m, \quad (5)$$

$$\sum_{j=1}^n \sum_{k=1}^{|\mathcal{P}|} t_{i,j} a_{k,j} x_{k,d} \leq \sum_{d_1=1}^d \sum_{d_2=d}^D y_{i,d_1,d_2}, \quad 1 \leq i \leq m, 1 \leq d \leq D. \quad (6)$$

Numerical Experiments

To find a numerical solution to the problem, we implemented Gurobi's Optimizer 13.0.1 interfaced with Python 3.12.8. In the case of ILP, the solver uses a **branch-and-cut** algorithm:

- **Solve** the LP relaxation with $x_{k,d}, y_{i,d_1,d_2} \in [0, 1]$
- If the solution is fractional, **cut** the feasible region without removing integer solutions
- If cuts are insufficient, **branch** into two subproblems by fixing a fractional variable to 0 or 1,
- **Discard** any subproblem with suboptimal LP bound
- **Repeat** until all subproblems are resolved and the optimal integer solution is found

Results and Observations

n	m	W	ρ	Feasibility	Cost	Time (in s)
10	20	50	0.3	Optimal	131014.0	5.652
10	10	75	0.3	Optimal	42162.0	0.488
10	10	50	0.5	Optimal	84293.0	0.882
10	10	50	0.3	Optimal	42162.0	0.415
15	20	75	0.5	Infeasible/Other	–	1.225
15	20	50	0.5	Infeasible/Other	–	0.756
15	10	50	0.5	Optimal	117608.0	25.010
15	10	50	0.3	Optimal	68596.0	17.311

From the experiments, it emerged that the ILP pattern model cannot handle a large number of scenes and actors. This scalability issue represents a critical limitation of the model and demonstrates the NP-hardness of the problem.

After running the program for 10.5 hours, we terminated it. We discovered that the case of $n = 20$, it took a large amount of time; this result was also found in [2].

Conclusion

We considered a talent scheduling problem, which takes into account daily capacity and changeover times. The problem was formulated as an application of integer linear programming and was solved numerically using a Gurobi optimizer interfaced with python. Possible future developments include:

- Allowing scenes to be shot on multiple days
- Allowing for actors to have days off even when their required scenes have yet to be completed
- Mapping the shooting days onto a calendar with forced holidays (e.g. Thanksgiving, Christmas, etc.)

References

- [1] Diamond J.E. Cheng T.C.E. and Lin B.M.T. Optimal scheduling in film production to minimize talent hold cost. *Journal of Optimization Theory and Applications*, 79:479–492, December 1993.
- [2] Zhaoqi Yang, Shunji Tanaka, and Bertrand M.T. Li. Talent scheduling with daily working capacity and scene changeover times. *Omega*, 138, March 2026.